QUANT SOCIETY PRESENTS

PYTHON CAMPUS CHALLENGE

PROBLEM SET 1 • CASE-BASED QUESTIONS

Azim Premji University, Bhopal

Python Session Series

INSTRUCTOR	Ramesh Beniwal
ROLE	Club Coordinator & Python Session Instructor
SESSIONS COVERED	Python Without Fear + Making Python Think
TOTAL QUESTIONS	16

INSTRUCTIONS

Attempt all questions. Write Python code using only the concepts covered in the two sessions: print, comments, variables, data types, numbers, float, string, boolean, arithmetic, input, comparison operators, logical operators, if-elif-else, for loop, range, lists, and filtering with for + if.

CHALLENGE MODE
Read each case carefully, identify the condition, and write clear, readable Python code.

PROBLEM SET 1 — QUESTIONS

Use the concepts from the sessions. Keep your code simple and add comments wherever asked.

QUESTIONS 1–8

Q1. Smart Q Discount Bill

You are ordering food from the Smart Q app in the cafeteria. Ask for the student's name, order amount, and loyalty points.

- If amount is greater than 200 and points are greater than 50, give 20% discount.
- Else if amount is greater than 100, give 10% discount.
- Else give no discount.

Print the final amount with proper comments.

Q2. Affordable Cafeteria Menu

The cafeteria menu is ["Samosa", "Poha", "Maggi", "Tea", "Cold Coffee"].

Prices are [20, 30, 60, 15, 80].

Use a for loop with range and if to print all items under ₹50 and count how many are affordable.

Q3. AI Assignment Deadline Checker

A student submits an AI assignment. Ask for submission hour and minute. Deadline is 18:00.

If time is invalid, print Invalid time.
If submitted before or at 18:00, print AI assignment submitted on time.
Otherwise print Late submission.

Q4. Marks Calculator with Grace

Ask for marks of 3 subjects. Calculate average.

If any subject is below 30, print Fail: one subject below 30.
Else if average ≥ 90 print Grade A,
else if average ≥ 75 print Grade B,
else if average ≥ 50 print Grade C,
else print Fail.

Q5. Ethical Friend Phone Prank Simulator

You want to make a funny Python prank on a friend's phone, but only with permission.

Ask: Do you have permission? yes/no.
If no, print Access denied. Ethical hacking needs consent.
If yes, ask friend's name and battery percent.
If battery < 20, print a funny low-battery message. Else print a safe prank message.

Q6. Variable Type Detective

Create these variables: token = 101, price = 45.5, food = "Poha", is_veg = True.

Print each value and its type. Then increase price by 10% and print the new value and type again.

Q7. Smart Q Delivery Speed

A Smart Q delivery person covers distances [1.2, 0.8, 2.5] km in times [5, 4, 10] minutes.

For each delivery, calculate speed in km/h. If speed > 15, print Fast delivery.

Q8. Swap Tokens

Create a = "SmartQ" and b = "Cafeteria". Swap them without using a third variable and print.

Also use multiple assignment to set x, y, z = 10, 20, 30 and print them.

PROBLEM SET 1 — QUESTIONS (CONTINUED)

Questions 9–16

Q9. Debug the Cafeteria Code

Find and correct the errors in this code:

```
1food = "Samosa" price = 20 if price = 20 print("Cheap") else: print("Expensive")
```

Q10. Free Dessert Condition

Write a condition for free dessert:

Free dessert if (is_veg and price > 150) or (day == "Friday" and has_coupon).

Test with is_veg = True, price = 180, day = "Monday", has_coupon = False.

Print whether the student gets free dessert.

Q11. Range Token Queue

Use range to print token numbers from 101 to 110. Skip token 105.

Then print only even token numbers from the same range, also skipping 105.

Q12. Filter AI Assignment Submissions

Given ["submitted", "late", "submitted", "missing", "submitted", "late"], use a for loop and if-elif-else to count how many are submitted, late, and missing. Print the summary.

Q13. Cafeteria Bill with Tax and Discount

Ask for total amount.

If total > 500, give 10% discount; otherwise no discount.

Then apply 5% tax on the amount after discount.

Print total, discount, tax, and final bill.

Q14. Boolean Confusion

Given A = True, B = False, C = True, evaluate and print:

1. not A and B or C
2. A or B and not C
3. (A and B) or (not B and C)

Q15. Smart Q Order Validator

Menu is ["Poha", "Samosa", "Maggi", "Tea"].

Ask for item and quantity. If item is in the menu and quantity > 0, print Order accepted; else print Invalid order. Use a for loop to check the item, not direct in.

Q16. Ethical Hacking Password Checker

Ask for a friend’s phone password in a simulated ethical hacking demo.

If password is "1234" or "password", print Weak: friend’s phone could be hacked. Change it.

Else if length is less than 8, print Medium weak: use at least 8 characters.

Else print Strong password.